TCO Welding Procedures Selection Spreadsheet for Pipe Classes specified in L-ST-2056 Rev 4 Rev. 2025_10

Piping Line Class	Design Code	Design Temp range, deg C	Pipe (as specified in line class)			Welding Process		Comments
			Material	Dia, in	Thickness range as per line class, mm	WPS #	PWHT Yes/No	
150A20	ASME B31.3	-40 / 200	Alloy 20	1/2 - 4	2.77 - 3.91	45AW	N	
150A80	ASME B31.3	-40 / 230	Incoloy 825 (NACE)	3/4 - 12	3.91 - 7.11	51AW	N	
150C80	ASME B31.3	-40 / 200	Incoloy 825	1/2-3 Solid	3.68 - 5.49	51AW	N	
			LTCS + 3 mm 625 internal cladding	4 - 16	9.27 - 11.13	81AW	N see Note (7)	
			LTCS + 3 mm 825 internal cladding	4 - 18	6.02 - 7.92	**Contact MI Engineer**	N see Note (7)	
			LTCS + 3 mm 825/625 internal cladding	18 - 48	9.53 - 12.70	**Contact MI Engineer**	N see Note (7)	
150CP2	ASME B31.3	-40 / 150	LTCS PTFE Lined	1/2 - 24		N/A	N/A	Flanged connections
150H21	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	3.73 - 7.82	20AW, for SW see Note (2)	N	SW only for legacy lines < 2"
				14 - 48	7.92 - 12.70	21AW	N	
				24	19.05	21PW	Y	
150H23	ASME B31.3	-40 / 215	LTCS NACE	1/2 - 3	7.47 - 11.07	20AW	N	
				> 3	8.56 - 12.70	21AW	N	
150H25 (A)	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	3.91 - 7.82	20PW	Y	For service codes CC, CCS, DCS
				> 3	7.92 - 12.70	21PW	Y	SW only for legacy lines < 2"
150H25 (B)	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	3.91 - 7.82	20AW, for SW see Note (2)	N	For service codes CC, CCS, DCS use 150H25 (A)
				> 3	7.92 - 12.70	21AW	N	SW only for legacy lines < 2"
150H26	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	3.91 - 5.54	20AW	N	
				> 3	6.02 - 10.00	21AW	N	
150H2G	ASME B31.3	-40 / 150	LTCS Galv/Scrd	< 3	5.54 - 7.14	N/A	N/A	Threaded connections
			LTCS Galv	≥ 3	7.62 - 9.53	21AW, Note (3)	N	Welded
150H5E	ASME B31.8	-40 / 75	LTCS	1/2 - 3	3.73 - 5.54	F-001AW, Note (6)	N	
			LTCS - API 5L X60 (SMLS)	6 - 8	6.35	F-002AW, Note (6)	N	
150J20	ASME B31.3	-40 / 180	LTCS NACE	3/4 - 3	5.49 - 8.74	20AW	N	
				> 3	6.02 - 9.53	21AW	N	
150K21	ASME B31.3	-40 / 232	LTCS NACE	1/2 - 3	3.73 - 8.74	20PW, for SW see Note (2)	Y	
				> 3	6.02 - 25.4	21PW	Y	
150K23	ASME B31.3	-40 / 200	LTCS NACE	3/4 - 3	7.62 - 11.07	20PW	Y	
				> 3	8.56 - 12.70	21PW	Y	
150K26	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	3.91 - 5.54	20PW, for SW see Note (2)	Y	SW only for legacy lines < 2"
				> 3	6.02 - 10.00	21PW	Y	
150K5E	ASME B31.8	-40 / 75	LTCS PTFE Lined			N/A	N/A	Flanged connections
150K5F	ASME B31.8	-40 / 100	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	3.91 - 8.74	F-001PW, Note (6)	Y	
			LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 6.35	F-002PW, Note (6)	Y	
				> 12	6.35 - 9.53	F-017PW, Note (6)	Y	
150M40	ASME B31.3	-40 / 215	Monel 400	3/4 - 2	2.87 - 3.91	48AW	N	
150MO6	ASME B31.3	-40 / 200	6% Moly (UNS31254)	1/2 - 3	3.91 - 5.08	01AW	N	SW only for legacy lines < 2"
				> 3	7.04 - 7.80	01AW	N	
150P21	ASME B31.3	-120 / 180	316/316L SS	1/2 - 3	3.91 - 7.82	25AW	N	
				> 3	7.11 - 24.61	14AW	N	
150P22	ASME B31.3	-40 / 535	316/316L SS	1/2 - 3	3.38 - 7.82	25AW	N	
				> 3	7.04 - 12.70	14AW	N	
150P23	ASME B31.3	-40 / 145	316 SS	1/2 - 2	3.91 - 5.54	25AW	N	For vendor packages only
150PE2		-40 / 40	HDPE			N/A	N/A	Thermal fusion technique
150PE3		-40 / 40	HDPE			N/A	N/A	Thermal fusion technique
150PE4		-40 / 40	HDPE			N/A	N/A	Thermal fusion technique
150PE7		-40 / 75	HDPE			N/A	N/A	Below ground thermal fusion technique
								Above ground flanged connections
150PE14		-40 / 40	HDPE			N/A	N/A	Thermal fusion technique

Piping Line Class	Design Code	Design Temp range, deg C	Pipe (as specified in line class)			Welding Process		Comments
			Material	Dia, in	Thickness range as per line class, mm	WPS #	PWHT Yes/No	
150PE15		-40 / 75	HDPE			N/A	N/A	Thermal fusion technique
150PE16		-40 / 75	HDPE			N/A	N/A	Thermal fusion technique
150VC01	ASME B31.3	-40 / 110	LTCS Galv	≤ 1		N/A	N/A	Screwed connections
				≥ 1.1/2 and ≤ 16		N/A	N/A	Victaulic groove & coupling
				> 16	7.92 - 9.53	21AW	N	
150VG01	ASME B31.3	-40 / 110	LTCS Galv	≤ 1		N/A	N/A	Screwed connections
				≥ 1.1/2 and ≤ 16		N/A	N/A	Victaulic groove & coupling
				> 16	7.92 - 9.53	21AW	N	
150VS01	ASME B31.3	-40 / 110	316/316L SS			N/A	N/A	Victaulic groove & coupling
300A20	ASME B31.3	-40 / 200	Alloy 20	1/2 - 3	3.91 - 5.56	45AW	N	
				> 3	7.11 - 9.52	46AW	N	
300A80	ASME B31.3	-40 / 230	Incoloy 825	1/2 - 12	3.91 - 9.52	51AW	N	
300C80	ASME B31.3	-40 / 230	Incoloy 825	1/2-3 Solid	3.68 - 5.49	51AW	N	
			LTCS + 3 mm 625 internal cladding	4-16	6.02 - 12.70	81AW	N see Note (7)	
			LTCS + 3 mm 825 internal cladding	4-16	6.02 - 12.70	**Contact MI Engineer**	N see Note (7)	
			LTCS + 3 mm 825/625 internal cladding	18 - 24	12.70 - 14.27	**Contact MI Engineer**	N see Note (7)	
			LTCS + 3 mm 825/625 internal cladding	42	24.00	**Contact MI Engineer**	Y	
300H21(A)	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	3.73 - 7.82	20AW, for SW see Note (2)	N	For service code CDM use 300H21 (B) SW only for legacy lines < 2"
				14 - 30	9.53 - 18.00	21AW	N	
				24 - 48	19.05 - 29.00	21PW	Y	
300H21 (B)	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	3.73 - 7.82	20PW, for SW see Note (2)	Y	For service code CDM only SW only for legacy lines < 2"
				14 - 30	9.53 - 18.00	21PW	Y	
				24 - 48	19.05 - 29.00	21PW	Y	
300H25 (A)	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	4.55 - 7.82	20AW, for SW see Note (2)	N	For legacy lines service code CC, CCS, DCS use 300H25 (B)
				> 3	8.18 - 17.48	21AW	N	SW only for legacy lines < 2"
300H25 (B)	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	4.55 - 7.82	20PW, for SW see Note (2)	Y	For legacy lines service codes CC, CCS, DCS only
				> 3	8.18 - 17.48	21PW	Y	SW only for legacy lines < 2"
300H2G	ASME B31.3	-40 / 150	LTCS Galv/Scrd	≤ 2	5.54 - 7.14	N/A	N/A	Threaded connections
				> 2	6.02 - 7.62	21AW, Note (3)	N	Welded
300H5F	ASME B31.8	-40 / 90	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	3.91 - 5.54	F-001AW, Note (6)	N	
			LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 7.09	F-002AW, Note (6)	N	
				> 12	7.92 - 9.53	F-017AW, Note (6)	N	
300H5P	ASME B31.8	-40 / 80	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	3.91 - 5.54	F-001AW, Note (6)	N	
			LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 6.35	F-002AW, Note (6)	N	
				> 12	6.35 - 9.53	F-017AW, Note (6)	N	
300J20	ASME B31.3	-40 / 180	LTCS NACE	3/4 - 3	3.91 - 5.56	20AW	N	
				> 3	6.35 - 9.53	21AW	N	
300K21	ASME B31.3	-40 / 260	LTCS NACE	1/2 - 3	4.55 - 7.82	20PW, for SW see Note (2)	Y	SW only for legacy lines < 2"
				> 3	8.18 - 36.0	21PW	Y	
300K23	ASME B31.3	-40 / 200	LTCS NACE	1/2 - 3	7.47 - 11.13	20PW	Y	
				> 3	10.97 - 20.62	21PW	Y	
300K5A (AG)	ASME B31.8	-40 / 88	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	4.55 - 7.47	F-001PW, Note (6)	Y	For above ground welds only
			LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 12.70	F-002PW, Note (6)	Y	
				> 12	7.14 - 34.93	F-017PW, Note (6)	Y	
300K5A (UG)	ASME B31.8	-40 / 88	LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 7.14	F-002AW, Note (6)	N	For below ground pipeline welds only
				> 12	7.14 - 10.31	F-017AW, Note (6)	N	
300K5D	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	4.55 - 7.47	20PW	Y	
				> 3	6.02 - 7.11	21PW	Y	
300K5E (AG)	ASME B31.8	-40 / 95	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	4.55 - 7.47	F-001PW, Note (6)	Y	For above ground welds only
			LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 7.14	F-002PW, Note (6)	Y	
				> 12	7.92 - 15.88	F-017PW, Note (6)	Y	

Piping Line Class	Design Code	Design Temp range, deg C	Pipe (as specified in line class)			Welding Process		Comments
			Material	Dia, in	Thickness range as per line class, mm	WPS #	PWHT Yes/No	
300K5E (UG)	ASME B31.8	-40 / 95	LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 7.14	F-002AW, Note (6)	N	For below ground pipeline welds only
				> 12	7.92 - 11.91	F-017AW, Note (6)	N	
300K5F (AG)	ASME B31.8	-40 / 80	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	3.91 - 5.54	F-001PW, Note (6)	Y	For above ground welds only
			LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 7.14	F-002PW, Note (6)	Y	
				> 12	7.92 - 11.91	F-017PW, Note (6)	Y	
300K5F (UG)	ASME B31.8	-40 / 80	LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 7.14	F-002AW, Note (6)	N	For below ground pipeline welds only
				> 12	7.92 - 11.91	F-017AW, Note (6)	N	
300K5G (AG)	ASME B31.8	-40 / 95	Incoloy 825	3/4-3 Solid	3.91 - 7.62	51AW	N see Note (7)	For above ground welds only
			LTCS - API 5L X60 (SMLS) + 3 mm 825 internal cladding	6-16	7.11 - 9.53	**Contact MI Engineer**	N see Note (7)	
300K5G (UG)	ASME B31.8	-40 / 95	Incoloy 825	3/4-3 Solid	3.91 - 7.62	51AW	N see Note (7)	For below ground pipeline welds only
			LTCS - API 5L X60 (SMLS) + 3 mm 825 internal cladding	6-16	7.11 - 9.53	**Contact MI Engineer**	N see Note (7)	
300MO6	ASME B31.3	-40 / 130	6% Moly (UNS31254)	1/2 - 3	3.91 - 5.49	01AW	N	SW only for legacy lines < 2"
300P21	ASME B31.3	-120 / 180	316/316L SS	1/2 - 3	3.91 - 5.08	25AW	N	SW only for legacy lines < 2"
				> 3	9.52 - 17.48	14AW	N	
300P22	ASME B31.3	-40 / 350	316/316L SS	1/2 - 3	3.91 - 7.82	25AW	N	
				> 3	9.52 - 17.48	14AW	N	
300PF2	ASME B31.3	-40 / 130	Glass Reinforced Epoxy			N/A	N	Adhesive taper joint
600A80	ASME B31.3	-40 / 230	Incoloy 825	1/2 - 6	3.68 - 10.97	51AW	N	
				8 -24	12.70 - 24.61	54AW		
				30 - 36	29.00 - 31.00	**Contact MI Engineer**		
			316/316L SS	30	38.00	14AW		
600C80	ASME B31.3	-40 / 200	Incoloy 825	1/2-3 Solid	3.68 - 5.56	51AW	N	
			LTCS + 3 mm 625 internal cladding	4 - 10	8.56 - 15.09	81AW	N see Note (7)	
			LTCS + 3 mm 625 internal cladding	16	21.44	**Contact MI Engineer**	Y	
			LTCS + 3 mm 825 internal cladding	4 - 10	8.56 - 15.09	**Contact MI Engineer**	N see Note (7)	
			LTCS + 3 mm 825 internal cladding	16	21.44	**Contact MI Engineer**	Y	
			LTCS + 3 mm 825/625 internal cladding	18-30	23.88 - 35.00	**Contact MI Engineer**	Y	
600H21	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	3.73 - 7.82	20AW, for SW see Note (2)	N	SW only for legacy lines < 2"
				4 - 10	8.56 - 15.09	21AW	N	
				16 - 30	21.44 - 34.00	21PW	Y	
600H25	ASME B31.3	-40 / 343	LTCS NACE	1/2 - 3	5.56 - 7.82	20AW	N	
				8 - 10	12.70 - 15.09	21AW	N	
				16 - 24	21.44 - 30.96	21PW	Y	
600H2G	ASME B31.3	-40 / 38	LTCS Galv/Scrd	≤ 2	5.56 - 8.74	N/A	N/A	Threaded connections
				> 2	11.13	20AW/21AW, Note (3)	N	
600H5M	ASME B31.8	-40 / 75	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	3.91 - 5.54	F-001AW, Note (6)	N	
			LTCS - API 5L X60 (SMLS)	4 - 8	6.02 - 7.11	F-002AW, Note (6)	N	
600H5P (AG)	ASME B31.8	-40 / 100	LTCS - SMYS≤ 42 ksi (290 Mpa)	1/2 - 3	3.91 - 7.82	F-001AW, Note (6)	N	For above ground welds only
			LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 11.13	F-002AW, Note (6)	N	
				> 12 - 18	11.13 - 11.13	F-017AW, Note (6)	N	
				≥ 24	14.27 - 15.88	F-017PW, Note (6)	Y	
600H5P (UG)	ASME B31.8	-40 / 100	LTCS - API 5L X60 (SMLS)	4 - 12	6.02 - 11.13	F-002AW, Note (6)	N	For below ground pipeline welds only
				≥ 24	14.27 - 14.27	F-017PW, Note (6)	Y	
600K21	ASME B31.3	-40 / 260	LTCS NACE	1/2 - 3	4.78 - 7.82	20PW, for SW see Note (2)	Y	SW only for legacy lines < 2"
				> 3	12.7 - 48.00	21PW	Y	
600K5D (AG)	ASME B31.8	-40 / 90	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	3.91 - 5.54	F-001PW, Note (6)	Y	For above ground welds only
			LTCS NACE - API X60 (SMLS)	4 - 12	5.16 - 8.38	F-002PW, Note (6)	Y	
				> 12	8.74 - 12.7	F-017PW, Note (6)	Y	
			LTCS NACE - API X60 (SAW)	28	14.27	F-003PW, F-008PW, Note (6)	Y	
			LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	3.91 - 5.54	F-001AW, Note (6)	N	

Piping Line Class	Design Code	Design Temp range, deg C	Pipe (as specified in line class)			Welding Process		Comments
			Material	Dia, in	Thickness range as per line class, mm	WPS #	PWHT Yes/No	
600K5D (UG)	ASME B31.8	-40 / 90	LTCS NACE - API X60 (SMLS)	4 - 12	5.16 - 8.38	F-002AW, Note (6)	N	For below ground pipeline welds only
				> 12	8.74 - 12.7	F-017AW, Note (6)	N	
			LTCS NACE - API X60 (SAW)	28	14.27	F-003AW, F-008AW, Note (6)	N	
600K5E (AG)	ASME B31.8	-40 / 88	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	1/2 - 3	5.56 - 8.74	F-001PW, Note (6)	Y	For above ground welds only
			LTCS NACE - API X60 (SMLS)	4 - 12	6.35 - 11.13	F-002PW, Note (6)	Y	
				> 12	11.91 - 20.62	F-017PW, Note (6)	Y	
600K5E (UG)	ASME B31.8	-40 / 88	LTCS NACE - API X60 (SMLS)	4 - 12	6.35 - 11.13	F-002AW, Note (6)	N	For below ground pipeline welds only UG, NPS 26" (3GI Feed Gas) with PWHT
				> 12	11.91 - 17.48	F-017AW, Note (6)	N	
				26	15.88 - 19.05	F-017PW, Note (6)	Y	
600K5F (AG)	ASME B31.8	-40 / 100	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	3.91 - 7.62	F-001PW, Note (6)	Y	For above ground welds only
			LTCS NACE - API X60 (SMLS)	4 - 12	6.02 - 10.31	F-002PW, Note (6)	Y	
				> 12	10.31 - 19.05	F-017PW, Note (6)	Y	
			LTCS NACE - API X60 (SAW)	28	19.05	F-003PW, F-008PW, Note (6)	Y	
600K5F (UG)	ASME B31.8	-40 / 100	LTCS NACE - API X60 (SMLS)	4 - 12	6.02 - 10.31	F-002AW, Note (6)	N	For below ground pipeline welds only
				> 12	10.31 - 15.88	F-017AW, Note (6)	N	
				24	19.05	F-017PW, Note (6)	Y	
			LTCS NACE - API X60 (SAW)	28	19.05	F-003PW, F-008PW, Note (6)	Y	
600K5G (AG)	ASME B31.8	-40 / 95	Incoloy 825	3/4-3 Solid	3.91 - 7.62	51AW	N see Note (7)	For above ground welds only
			LTCS - API 5L X60 (SMLS) + 3 mm 825 internal cladding	6-16	7.11 - 12.70	**Contact MI Engineer**	N see Note (7)	
600K5G (UG)	ASME B31.8	-40 / 95	Incoloy 825	3/4-3 Solid	3.91 - 7.62	51AW	N see Note (7)	For below ground pipeline welds only
			LTCS - API 5L X60 (SMLS) + 3 mm 825 internal cladding	6-16	7.11 - 12.70	**Contact MI Engineer**	N see Note (7)	
600K5H (AG)	ASME B31.8	-40 / 95	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	1/2 - 3	5.56 - 8.74	F-001PW, Note (6)	Y	For above ground welds only
			LTCS NACE - API X60 (SMLS)	4 - 12	6.35 - 18.26	F-002PW, Note (6)	Y	
				> 12	14.27 - 20.62	F-017PW, Note (6)	Y	
600K5H (UG)	ASME B31.8	-40 / 95	LTCS NACE - API X60 (SMLS)	4 - 12	6.35 - 12.7	F-002AW, Note (6)	N	For below ground pipeline welds only
				> 12	14.27 - 17.48	F-017AW, Note (6)	N	
				≥ 20	17.48 - 20.62	F-017PW, Note (6)	Y	
600P21	ASME B31.3	-120 / 180	316/316L SS	1/2 - 3	3.73 - 5.56	25AW	N	SW only for legacy lines < 2"
				> 3	12.70 - 50.0	14AW	N	
600P22	ASME B31.3	-46 / 350	316/316L SS	1/2 - 3	3.73 - 7.82	25AW	N	
				> 3	12.70 - 50.0	14AW	N	
900A80	ASME B31.3	-40 / 300	Incoloy 825	1/2 - 6	3.68 - 10.97	51AW	N	
				8 - 24	12.70 - 24.61	54AW		
				30	30.00	**Contact MI Engineer**		
			316/316L SS	2 - 4	5.54 - 11.13	25AW/14AW		
				24	46.02	14AW		
900C80	ASME B31.3	-40 / 220	Incoloy 825	3/4-3 Solid	3.91 - 7.62	51AW	N	
			LTCS + 3 mm 625 internal cladding	4 - 8	11.13 - 15.09	81AW	N see Note (7)	
			LTCS + 3 mm 625 internal cladding	16	26.19	**Contact MI Engineer**	Y	
			LTCS + 3 mm 825/625 internal cladding	18-36	29.36 - 57.00	**Contact MI Engineer**	Y	
900H21	ASME B31.3	-40 / 410	LTCS NACE	3/4 - 3	3.91 - 7.82	20AW, for SW see Note (2)	N	SW only for legacy lines < 2"
				4 - 4	11.13 - 11.13	21AW	N	
				16 - 30	30.96 - 49.00	21PW	Y	
900H25	ASME B31.3	-40 / 410	LTCS NACE	3/4 - 3	5.56 - 11.13	20AW	N	
				4 - 4	11.13 - 11.13	21AW	N	
				16 - 24	30.96 - 52.37	21PW	Y	

Piping Line Class	Design Code	Design Temp range, deg C	Pipe (as specified in line class)			Welding Process		Comments
			Material	Dia, in	Thickness range as per line class, mm	WPS #	PWHT Yes/No	
900H5H (AG)	ASME B31.8	-40 / 80	LTCS - SMYS≤ 42 ksi (290 Mpa)	1/2 - 3	5.56 - 11.13	F-001AW, Note (6)	N	For above ground welds only
			LTCS - API X60 (SMLS)	4 - 6	6.02 - 7.11	F-002AW, Note (6)	N	
900H5H (UG)	ASME B31.8	-40 / 80	LTCS - API X52 (SMLS)	2	6.00	F-001AW, Note (6)	N	For below ground pipeline welds only
			LTCS - API X60 (SMLS)	4 - 6	6.02 - 6.35	F-002AW, Note (6)	N	
900K21	ASME B31.3	-40 / 260	LTCS NACE	1/2 - 3	4.78 - 11.13	20PW	Y	
				> 3	11.13 - 59.0	21PW	Y	
900K23	ASME B31.3	-46 / 200	LTCS NACE	1 - 3	9.09 - 11.07	20PW	Y	
				> 3	17.12 - 36.53	21PW	Y	
900K5B (AG)	ASME B31.8	-40 / 90	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	5.56 - 11.13	F-001PW, Note (6)	Y	For above ground welds only
			LTCS NACE - API X60 (SMLS)	4 - 12	8.56 - 15.88	F-002PW, Note (6)	Y	
				> 12	19.05 - 19.05	F-017PW, Note (6)	Y	
900K5B (UG)	ASME B31.8	-40 / 90	LTCS NACE - API X60 (SMLS)	≥ 6	7.92 - 7.92	F-002AW, Note (6)	N	For below ground pipeline welds only
900K5C (AG)	ASME B31.8	-40 / 90	LTCS NACE - API X60 (SMLS)	6 - 8	9.53 - 10.31	F-002PW, Note (6)	Y	For above ground welds only
900K5C (UG)	ASME B31.8	-40 / 90	LTCS NACE - API X60 (SMLS)	6 - 8	8.74 - 9.53	F-002AW, Note (6)	N	For below ground pipeline welds only
900K5D (AG)	ASME B31.8	-40 / 90	LTCS NACE - SMYS≤ 42 ksi (290 Mpa)	3/4 - 3	5.56 - 11.13	F-001PW, Note (6)	Y	For above ground welds only
			LTCS NACE - API X60 (SMLS)	4 - 12	8.56 - 15.88	F-002PW, Note (6)	Y	
				> 12	19.05 - 19.05	F-017PW, Note (6)	Y	
900K5D (UG)	ASME B31.8	-40 / 90	LTCS NACE - API X60 (SMLS)	≥ 6	7.92 - 11.10	F-002AW, Note (6)	N	For below ground pipeline welds only
900P21	ASME B31.3	-140 / 350	316/316L SS	1/2 - 3	3.73 - 7.62	25AW	N	SW only for legacy lines < 2"
				> 3	11.13 - 56.00	14AW	N	
900P22	ASME B31.3	-40 / 350	316/316L SS	3/4 - 3	3.91 - 7.82	25AW	N	
				> 3	11.13 - 56.00	14AW	N	
1500A60	ASME B31.3	-100 / 110	Inconel 625 (NACE)	3/4 - 6	3.68 - 10.97	43AW	N	
				> 6	12.70	42AW	N	
1500F22	ASME B31.3	-40 / 155	LTCS NACE	3/4 - 1 1/2	7.82 - 10.15	20PW	Y	
			2-1/4Cr F22 Modified	≤ 4	7.00 - 10.00	HP700PW	Y	
				6 - 8	13.00 - 16.00	**Contact MI Engineer**	Y	
				10 - 12	19.00 - 22.00	**Contact MI Engineer**	Y	
				> 14	24.00- 47.50	**Contact MI Engineer**	Y	
1500H1	ASME B31.3	-40 / 343	LTCS NACE	3/4 - 3	5.56 - 8.74	20AW, for SW see Note (2)	N	SW only for legacy lines < 2"
				4 - 4	13.49 - 13.49	21AW	N	
				≥ 6	21.95 - 34.00	21PW	Y	
1500K1	ASME B31.3	-40 / 126	LTCS NACE	3/4 - 3	7.82 - 11.07	20PW	Y	
				4 - 20	17.12 - 54.00	21PW	Y	
1500P1	ASME B31.3	-50 / 200	316/316L SS	3/4 - 3	5.56 - 8.74	25AW, for SW see Note (2)	N	SW only for legacy lines < 2"
				> 3	17.12 - 18.26	14AW	N	
1500P2	ASME B31.3	-100 / 200	316/316L SS	3/4 - 3	5.56 - 8.74	25AW	N	
				> 3	17.12 - 17.12	14AW	N	
2500A60	ASME B31.3	-100 / 180	Inconel 625 (NACE)	3/4 - 4	3.91 - 8.56	43AW	N	
				≥ 4	11.13 - 14.27	42AW	N	
2500F22	ASME B31.3	-40 / 180	LTCS NACE	3/4 - 1 1/2	7.82 - 11.00	20PW	Y	
			2-1/4Cr F22 Modified	≤ 4	9.00 - 14.00	HP700PW	Y	
				6 - 8	19.50 - 24.00	**Contact MI Engineer**	Y	
				10 - 12	29.00 - 34.00	**Contact MI Engineer**	Y	
				> 14	37.00 - 66.00	**Contact MI Engineer**	Y	
2500H1	ASME B31.3	-40 / 200	LTCS NACE	3/4 - 2	7.47 - 12.00	20AW	N	
				3 - 3	16.00 - 16.00	21AW	N	
				≥ 4	20.00 - 66.00	21PW	Y	
2500K1	ASME B31.3	-40 / 200	LTCS NACE	3/4 - 3	7.82 - 16.00	20PW	Y	
				> 3	20.00 - 29.00	21PW	Y	
2500P1	ASME B31.3	-40 / 120	316/316L SS	3/4 - 1 1/2	7.82 - 10.15	25AW	N	

Piping Line Class	Design Code	Design Temp range, deg C	Pipe (as specified in line class)			Welding Process		Comments
			Material	Dia, in	Thickness range as per line class, mm	WPS #	PWHT Yes/No	
2500P1	ASME B31.3	-40 / 120	316/316L SS	≥ 2	12.70 - 22.00	14AW	N	
2500P2	ASME B31.3	-135 / 120	316/316L SS (F321 Flange)	3/4 - 1 1/2	7.82 - 10.15	25AW	N	
				≥ 2	12.70 - 24.00	14AW	N	
2500P3	ASME B31.3	-135 / 120	316/316L SS (F321 Flange)	3/4 - 1 1/2	7.82 - 10.15	25AW	N	
				≥ 2	12.70 - 22.00	14AW	N	
10000A60	ASME B31.3 (Ch.IX) API 6A	-120 / 130	Inconel 625 (NACE)	3/4 - 2	5.56 - 8.74	43AW	N	
				3 - 6	11.13 - 20.00	42AW	N	
				8	26.00	Contact MI Engineer	N	
				16	47.00	Contact MI Engineer	N	
10000F22	ASME B31.3 (Ch.IX) API 6A	-40 / 130	LTCS NACE	3/4 - 1 1/2	7.82 - 12.00	20PW	Y	
			2-1/4Cr F22 Modified	≤ 4	10.50 - 17.00	HP700PW	Y	
				6 - 8	23.50 - 29.50	Contact MI Engineer	Y	
				10 - 12	36.00 - 42.00	Contact MI Engineer	Y	
				> 14	46.00- 70.50	Contact MI Engineer	Y	
10000H5A	ASME B31.8 API 6A	-40 / 110	API 5L X60 (SMLS)	2 - 3	11.07 - 17.48	F-002PW	Y	
				4	19.51	F-002PW	Y	
10000K5A	ASME B31.8 API 6A	-40 / 90	API 5L X60 (SMLS)	2 - 3	11.07 - 17.48	F-002PW	Y	
				4	19.51	F-002PW	Y	
10000X1	ASME B31.3 (Ch.IX) API 6A	0 / 110	API 5L X60 (SMLS)	14	55.00	Contact MI Engineer	Y	
10000X22	ASME B31.8 API 6A	-40 / 110	2-1/4Cr F22 Modified	14	55.00	Contact MI Engineer	Y	

NOTES

(1) 3/4" & smaller -- Tube compression fittings

(2) Where possible butt weld joints to be considered to minimize number of SW. For SW welding and NDE requirements use TCO Socket Welding Guideline For Process Piping.

(3) All continuously wetted systems shall not be galvanized. All wet/dry systems and WD/PLM services are to be galvanized after fabrication

(4) Socket weld joints should not be used for corrosive or stress cracking services such as acids, amine, caustic, chlorides etc.

(5) All threaded connections except on orifice flanges shall be seal welded. All threaded connections on orifice flanges shall be bridge welded.

(6) WPS for branch connections (Olets) welding shall be selected according to NPS of branch, contact Materials Engineer if clarifications are required.

(7) Using a welding procedure without a PWHT. Deviation from the requirements of specifications L-ST-2056 Rev.U4 and PIM-SU-5112-TCO Rev.U05

GENERAL NOTES

Materials not listed in piping class shall only be used with approval of Mechanical Integrity (MI) Group. Special WPS to be developed and qualified for unlisted materials.

This table cannot be used as a reference table for design temperature ratings for line classes

Check wall thickness for non-standard schedules to ensure WPS / PQR are qualified.

Dual certified 316/316L stainless steel (i.e. chemical composition of 316L plus mechanical properties of 316).

For the line classes not listed in this table refer to the Legacy KTL, SGP / SGI & Field WPS spreadsheet. Contact MI Group if line class is not listed in both tables.

For NDE requirements refer to NDE table from RE QM Group